

Product Requirement Document (PRD) - Personalized Recommendation Widgets for OTTplay Users

1. Overview

1.1 Product Name

Personalized Recommendation Widgets

1.2 Product Description

This feature introduces personalized widgets on the OTTplay app, TV, Tablet, and web platforms. These widgets will display tailored content recommendations based on users' activities, enhancing their experience by suggesting relevant movies, shows, and web series.

The widgets will be driven by five key user actions:

- **Because You Watched:** Recommends content based on recently watched titles.
- **Because You Liked:** Displays suggestions based on content the user has liked.
- **Because You Added to Watchlist:** Suggests content based on items added to the user's watchlist.
- **Because You Downloaded:** Offers recommendations tied to downloaded content.
- **Because You Searched:** Suggests content similar to what the user has searched for.

2. Objectives

- Increase user engagement by delivering highly relevant, personalized content recommendations.
- Boost average watch time per user through improved content discovery.
- Drive higher OTT data consumption by encouraging users to explore suggested content.
- Ensure fresh and dynamic content recommendations by excluding completed and ongoing content.

3. Target Audience

3.1 User Persona

- **Who:** All logged-in users of OTTplay.com (App, TV, Tablet, and Web).
- **Needs:** Quick, relevant content suggestions that align with their preferences and past interactions on the OTTplay platform.

4. Feature Requirements

4.1 Core Features

Widget Categories

- **Because You Watched:** Suggests content based on recently watched titles ($\geq 50\%$, keep it customizable as we will keep changing it based on average completion rate).
 - Maximum 5 widgets
- **Because You Liked:** Recommends content based on user's liked titles.
 - Timestamp to be captured
 - Unlike: The moment a user unlike the same title, please remove it from consideration
 - Maximum 4 widgets
- **Because You Added to Watchlist:** Displays suggestions similar to the titles in a user's watchlist.
 - Timestamp to be captured
 - Removed from Watchlist: The moment a user removed the same title from watchlist, please remove it from consideration
 - Maximum 3 widgets
- **Because You Downloaded:** Provides recommendations tied to previously downloaded content.
 - Timestamp to be captured
 - Deleted from Downloads: The moment a user removed the same title from downloads, please remove it from consideration
 - If the user has watched the downloaded title more than 50% (keep it customizable as we will keep changing it based on average completion rate) but it is not getting considered under "Because you watched" then consider it for providing recommendation under "Because you Downloaded"

- Maximum 2 widgets
- **Because You Searched:** Offers content similar to past search queries.
 - Open Pointer: Jitendra is working on it
 - Only 1 Widget

Exclusions (Specific to "Because You Watched")

- Do not recommend content marked as "Watched" or "Watch Again".
- Exclude titles that appear in "Continue Watching".

4.2 Technical Requirements

Data Inputs

- **User ID:** Unique identifier for each user.
- **Search History:** Search query string, search suggestion clicks, search results title page clicks, timestamp.
- **Watchlist:** Title ID & timestamp of addition.
- **Likes:** Title ID & timestamp of like.
- **Downloads:** Title ID & timestamp of download.
- **Watch History:** Titles watched, watch duration (≥ 5 minutes but $< 95\%$).
- **Content Metadata:** Title ID, Genre, Cast, Director, Theme/Mood, Popularity Score.

Recommendation Engine

- Leverage content metadata to determine similarity.
- Implement caching strategies to optimize performance.
- Update recommendations in Real time.

Scalability & Performance Optimization

- **Cache Duration:** 1 hour (for "Because You Watched" widget) and 12 hours (for other widgets). Tentative, to be decided.
- **Sorting:** Prioritize most recent actions.
- **Database Query Optimization:** Minimize direct DB queries through caching.
- **Upper Limit:** Cap at 50 recommendations per widget (keep it customizable)

4.3 Negative Scenarios

- If a user has liked something and then unliked it, it will be removed from the “Liked” list and that particular title will not be considered for personalised widget
- If multiple actions are taken, they should be ranked accordingly:
 - **Watched** - Rank 1 (highest)
 - **Liked** - Rank 2
 - **Downloaded** - Rank 3
 - **Watchlist** - Rank 4
 - **Searched** - Rank 5 (lowest)
- Any content that a user has disliked should be completely removed from recommendations.

5. User Stories

1. As a logged-in OTTplay user, I want to see content suggestions based on my recent searches, so I can quickly find similar movies or shows.
2. As a logged-in OTTplay user, I want recommendations based on my watchlist, so I can explore related content.
3. As a logged-in OTTplay user, I want to see suggestions tied to content I’ve liked, so I can discover more content aligned with my preferences.
4. As a logged-in OTTplay user, I want recommendations based on my downloads, so I can find additional offline-viewing options.
5. As a logged-in OTTplay user, I want to receive recommendations based on my recently watched content while excluding completed and ongoing titles.

6. Success Metrics

6.1 Key Performance Indicators (KPIs)

- **User Engagement:** Increase CTR on recommended content by X% within 3 months.
- **Watch Time:** X% increase in average watch time per user within 3 months.
- **OTT Partner Consumption:** X% improvement in playing DAU% within 3 months.
- **Widget Interaction:** X% of logged-in users interact with at least one personalized widget per week.

6.2 Additional Metrics

- **Time spent on the platform per session.**
- **Conversion rate of recommendations to watchlist additions or downloads.**

7. Dependencies

- **User Subscription System:** Determines user eligibility for recommendations.
- **Watch History Database:** Required for fetching "Because You Watched" recommendations.
- **Similar Recommendations Collection:** Used for content suggestions.
- **Caching Layer:** Enhances response time and reduces database load.
- **Backend Team:** Builds or enhances the recommendation engine and API endpoints.
- **Frontend Team:** Integrates widgets into the app, TV, and web interfaces.
- **Analytics Team:** Ensures accurate user activity tracking.

8. Open Questions

1. **Widget Refresh Frequency:** Suggest 12-hour refresh cycle; real-time updates may be explored later.
2. **Upper Limit on Recommendations:** Suggest a cap of 50 recommendations per widget.
3. **A/B Testing for Widget Placement:** Requires UX research to optimize user engagement.

9. Future Enhancements

- **User Preference Learning:** Improve recommendations by analyzing watch duration trends.
- **Genre-Based Filtering:** Personalize recommendations based on preferred genres.
- **Multi-Device Sync:** Ensure recommendations align across multiple devices.

10. Industry Reference

How OTT Platforms Generate Personalized Recommendations

1. *"Because You Searched"*

- **Approach:** Keyword matching, genre classification, popularity-based ranking.

2. *"Because You Added to Watchlist"*

- **Approach:** Content similarity analysis, collaborative filtering based on other users with similar watchlists.

3. *"Because You Liked"*

- **Approach:** Identifying content themes, director/actor-based clustering, and user behavior modeling.

4. *"Because You Downloaded"*

- **Approach:** Offline-friendly content recommendations, focusing on frequently downloaded genres.

5. *"Because You Watched"*

- **Approach:** Exclude completed content, prioritize recently watched, and match similar themes.

This PRD ensures that personalized widgets cater to multiple user actions while optimizing recommendation accuracy and performance.